CORTEVA™ agriscience

Codex Monday Board Data Extraction and Reporting

Status Update / Demo

Last Updated: 7/28/2025

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Goals

1

Extract and store data from Monday.com

2

Enable historical tracking and schema evolution handling

3

Create Power BI report for visibility and self-service updates

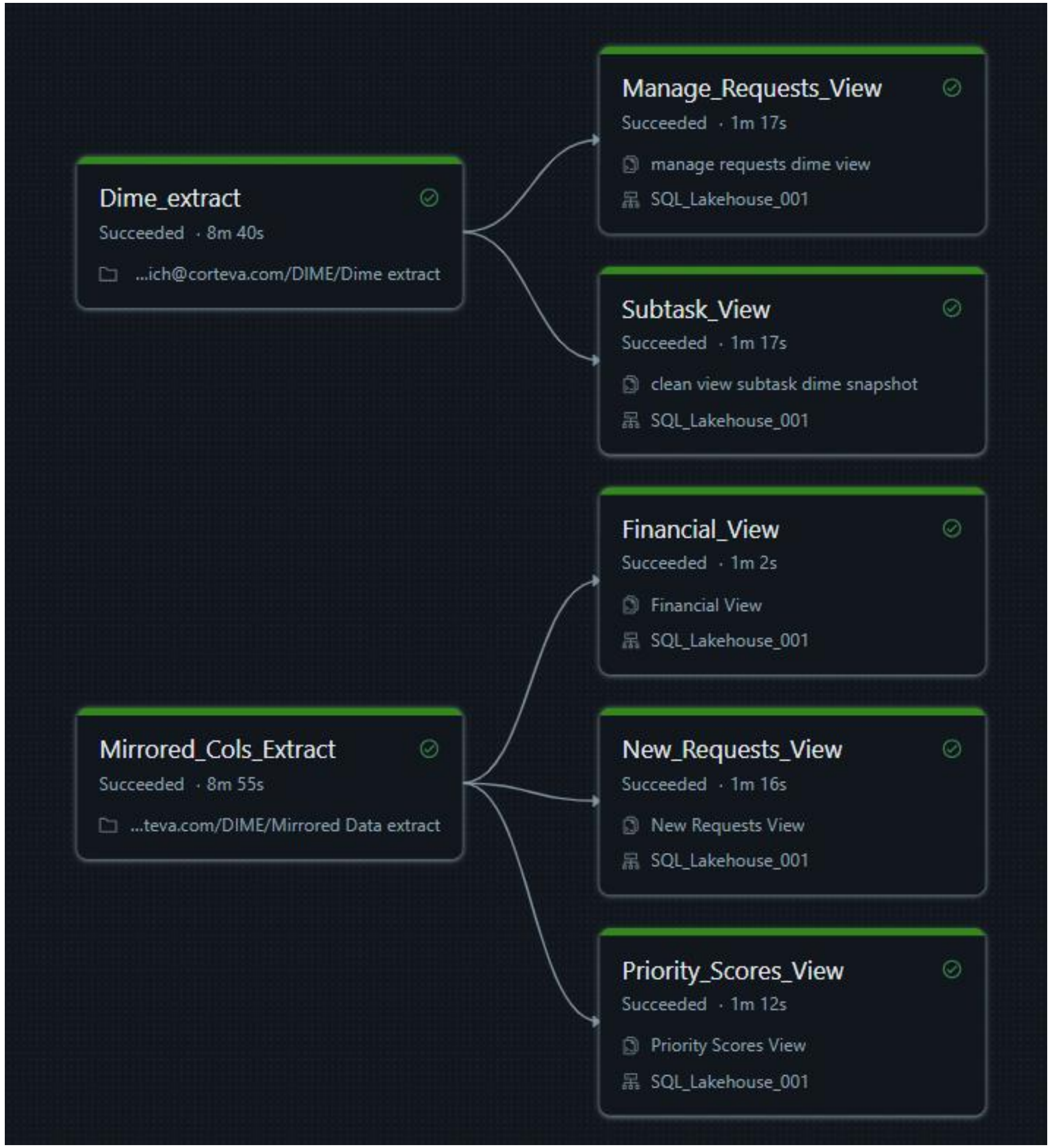
Architecture Overview

Overview:

- Data extraction via API Call (design and challenges)
- Pagination
- Subtasks
- Null or Empty Columns
- Dynamic Vs Stable boards (schema evolution)

- Medallion architecture
- Bronze Table: Raw Data (DLT for stable boards , Parquet files in a volume for Dynamic boards.)
- Silver/Gold: Transformed data (views/tables) connected to Power BI.

Databricks Job Run



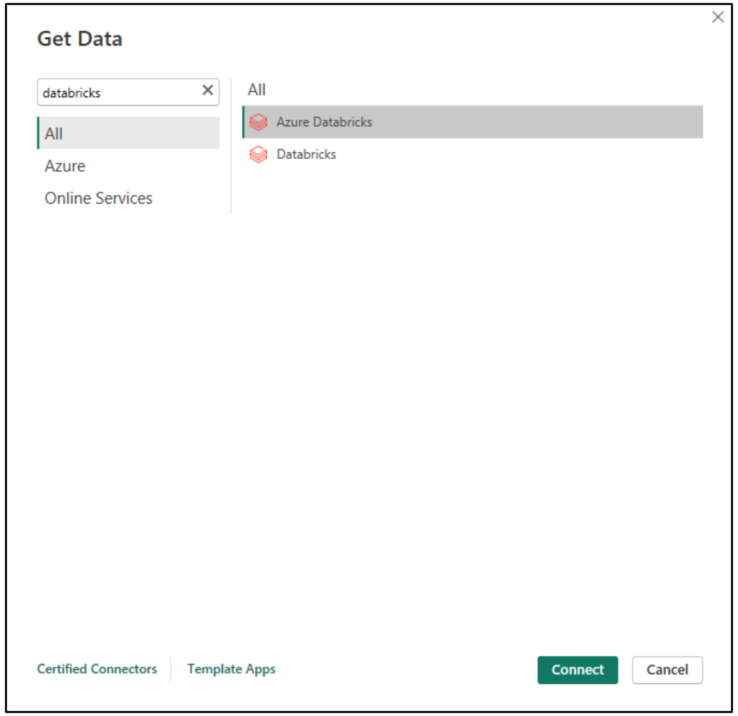
Databricks Asset Bundles (DABs)

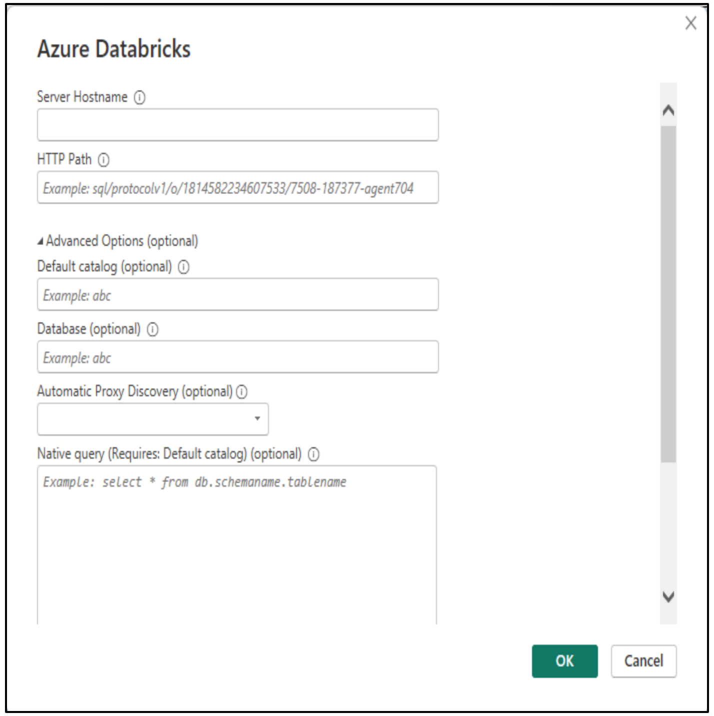
When you edit in Databricks UI, there's **no record** of what changed, when, or why. It is also run with code on my personal machine with my credentials (which will go away when my internship is complete)

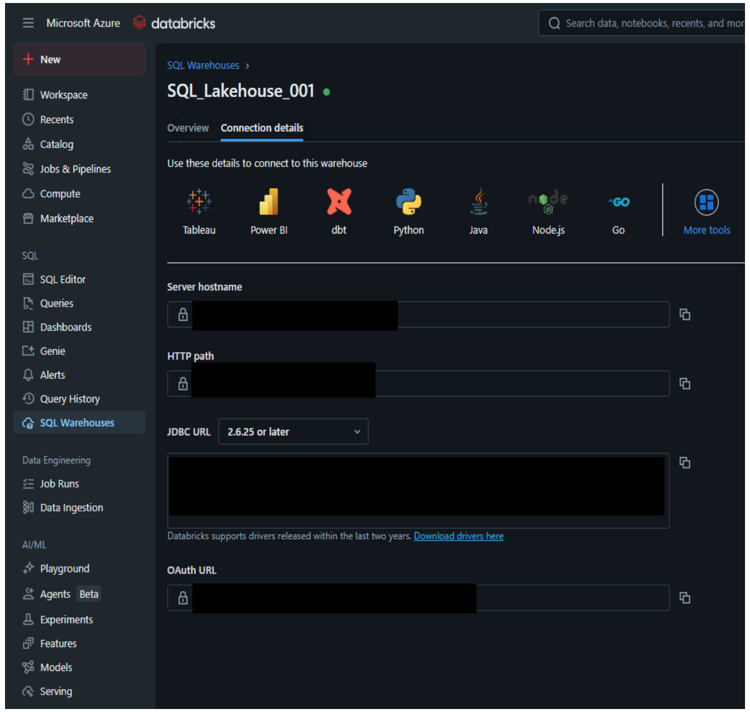
With DABs + Git in DevOps, every change is:

- Tracked in **version control**
- Tied to a **commit**
- Can be **rolled back** if it breaks
- Reviewed in **pull requests** by your team
- Parameterized (like dev, test, prod)

Data Import Process Databricks → Power BI







Database Relation in Power BI

The Monday.com API assigns a MondayID to every row that you pull

- These do not tie together records that span multiple boards, they are all distinct to each board
- In Monday.com you can easily see that they are linked / related, but nothing in the back-end shows this relationship

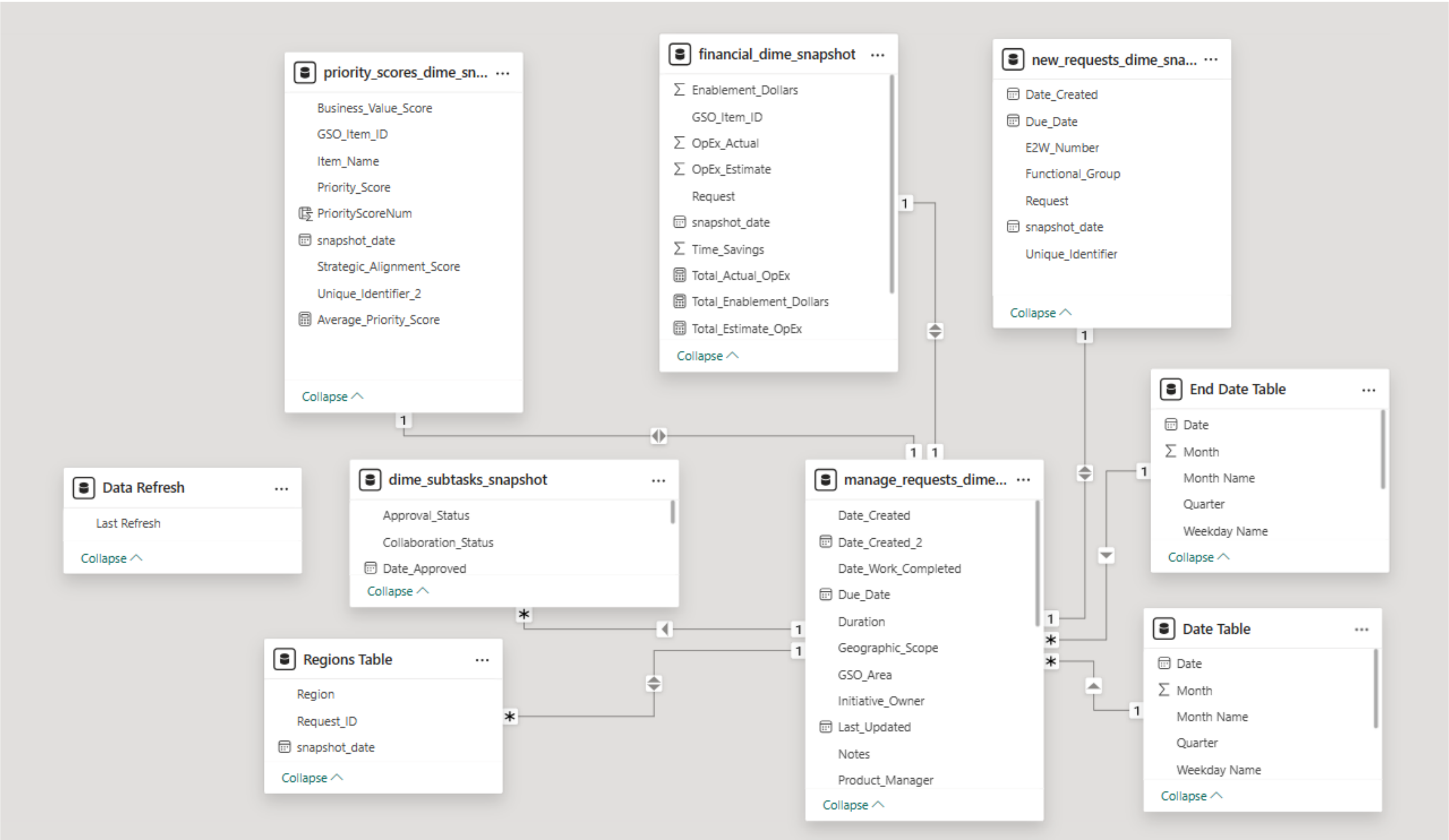
You must manually create your own system of tying together records

- GSO ID, Parent Monday ID, distinct names

I found that doing this process in Power BI was much easier than Databricks because of the visibility

- Drag and drop, error messaging, access to the data / queries at the same time

Database Relation in Power BI



Data Transformation in Power BI

The Monday.com API automatically pulls in all data as a text datatype

- This means financial values or dates can not be interpreted as what they are

I found that manipulating data in Power BI was much easier than in Databricks

- There is a simple UI where you can see the datatype of all columns, takes only the push of a button to manipulate instead of lines of code
- You can complete calculations (average, max, min, sum) without having to add it as a variable in SQL
- Easy paper trail where you can undo and see the changes you have made
- One change stays permanently, rather than having to re-run that code every day you update the data

Data Transformation DAX Columns

Raw_Timeline	In_Refinement_Timeline	Refined_Timeline
2025-06-01 - 2025-06-22	2025-06-01 - 2025-09-30	2025-08-01 - 2025-09-30

```
Raw_Start =
VAR RawTimeline = TRIM(current_lakehouse_snapshot[Raw_Timeline])
VAR StartText = LEFT(RawTimeline, 10)
RETURN
    IF(
        RawTimeline = "" || ISBLANK(RawTimeline),
        BLANK(),
        DATEVALUE(StartText)
    )
```

Current_Phase	Raw_Start	Raw_End	In_Refine_Start	In_Refine_End	Refined_Start	Refined_End
Not Started	6/1/2025	6/22/2025	6/1/2025	9/30/2025	8/1/2025	9/30/2025

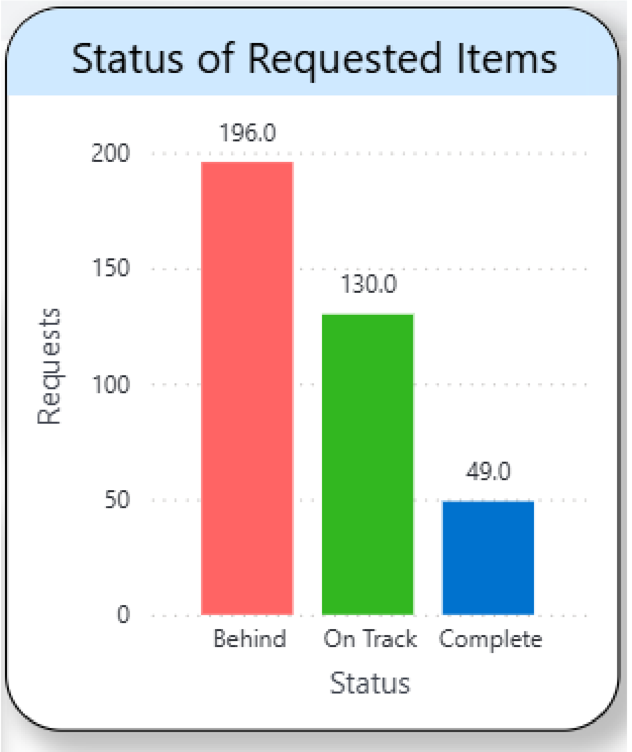
```
PlannedPhase =
IF (
    TRIM(current_lakehouse_snapshot[Raw_Timeline]) = ""
    || TRIM(current_lakehouse_snapshot[In_Refinement_Timeline]) = "",

    current_lakehouse_snapshot[Current_Phase],

    SWITCH(
        TRUE(),
        TODAY() >= [Raw_Start] && TODAY() <= [Raw_End], "Raw",
        TODAY() >= [In_Refine_Start] && TODAY() <= [In_Refine_End], "In Refinement",
        TODAY() >= [Refined_Start] && TODAY() <= [Refined_End], "Refined",
        TODAY() < [Raw_Start], "Not Started",
        TODAY() > [Refined_End], "Refined",
        "Out of Scope"
    )
)
```

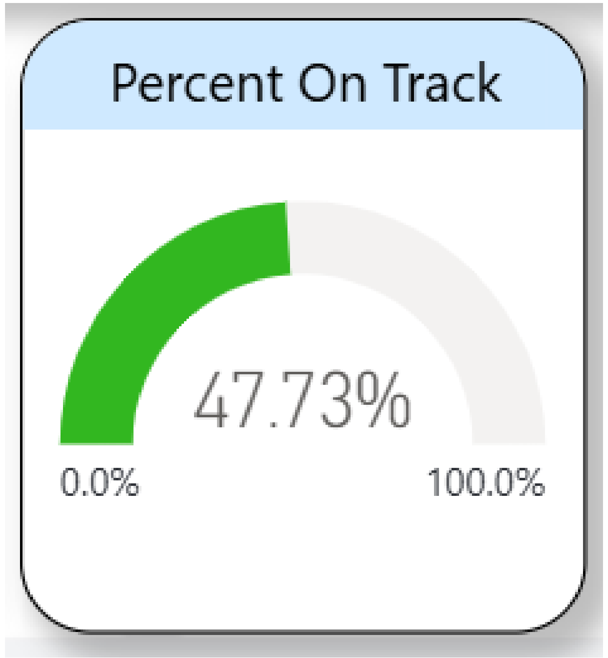
PlannedPhase
In Refinement

```
TrackStatus =
VAR Planned =
    SWITCH([PlannedPhase],
        "Not Started", 0,
        "Raw", 1,
        "In Refinement", 2,
        "Refined", 3,
        -1
    )

VAR Actual =
    SWITCH([Current_Phase],
        "Not Started", 0,
        "Raw", 1,
        "In Refinement", 2,
        "Refined", 3,
        -1
    )

RETURN
    SWITCH(
        TRUE(),
        Actual = 3 && TODAY() > [Refined_End], "Complete",
        Actual = Planned, "On Track",
        Actual > Planned, "Ahead",
        Actual < Planned, "Behind",
        "Unknown"
    )
```

TrackStatus
Behind

Data Transformation DAX Columns





Dashboard Demo